

Analysis

CAUTION: Tentative (Expected Effects are not detailed enough, just roughly written)

1) Feature description — Describe each new feature you added:

#	Feature Name	Design Rationale	Expected Effect (which entity types?)
1	<code>word.shape</code>	Captures capitalization and digit patterns (e.g., 'London' → 'Aaaaaa', '2024' → 'dddd').	PER, ORG (abbreviations)
2-5	Extended Affixes (<code>[-4:]</code> , <code>[:2~4]</code>)	Identifies distinct English morphemes (prefixes/suffixes) like <code>-land</code> or <code>Inter-</code> that strongly correlate with proper nouns.	ORG, LOC
6	<code>word.has_hyphen</code>	Punctuation indicator to capture compound or hyphenated proper nouns like 'New-York' or 'Coca-Cola'.	ORG, LOC, MISC
7	<code>word.has_dot</code>	Punctuation indicator to cleanly capture acronyms and abbreviations like 'U.S.'.	ORG, LOC
8	<code>word.is_alnum</code>	Distinguishes between pure text/number tokens and purely symbolic/punctuation tokens.	All (filtering out 'O' tokens)
9	<code>word.length</code>	Word length information helps filter out excessively short particles or unusually long non-entity strings.	All entities
10-11	POS Bigrams (<code>-1</code> , <code>+1</code> , <code>postag_bigram</code>)	Captures surrounding syntactic patterns like 'Adjective + Noun' that commonly bracket entities, providing stronger boundary context.	All (resolving B/I confusions)
12-14	Extended context (<code>-2:word</code> , <code>postag</code>)	Looks two words backward to capture preceding contextual triggers like 'Former President [NAME]'.	PER, ORG
15-17	Extended context (<code>+2:word</code> , <code>postag</code>)	Looks two words forward to capture succeeding indicators, such as specific verbs or positional titles.	PER, ORG
18	<code>isdigit_and_next_is_title</code>	Helps identify cases like "1995 World Cup" where a number is directly part of an event name.	MISC
19	<code>sentence.is_headline</code>	Flags if > 50% of the sentence consists of uppercase words, adjusting for contexts where normal <code>istitle()</code> rules fail.	All (robustness in All-Caps sentences)
20-31	Wide Context Windows (<code>-3</code> , <code>+3</code> , <code>-4</code> , <code>+4</code>)	Provides a broader view to capture strong semantic triggers (e.g., 'said', 'League') that are far away from the entity token.	All (resolving long-distance dependencies)

Additional Comments:

- Context Extension Features (12~17, 20~31)
- Affix Extension Features (2-5)
- Number Followed by Title Feature (18)
- Headline Feature (19)

⇒ These are directly originated from the error analysis and improvement ideas from Part1-Q3